

YenCare Health Platform

Week 3 Task Status & Master Sprint Execution Report (Gate 3 Feature-Complete)

Programme / Context	AmaliTech CSR Capstone Internship Programme — Product 4 (KNUST Health Services)
Active Sprint State	Gate 2 100% Verified & Done • Week 3 Major Backend Engine Milestones Merged into main
Current Execution Date	Tuesday, 15 September 2026 (Week 3 Execution / 72h to Gate 3 Feature Freeze)
Milestone Objective	Deliver complete patient virtual queue tracking and cancellation flows; achieve 100% Feature-Complete by Friday 18 Sep 2026.
Automated Test Health	108 Passing Tests across 31 Suites (0 Failures, Native Node Test Runner)
Lead Author & Role	Blessing Edmund Kwame Dogbe (Backend & DevOps Lead / Product Coordinator)

1. Team Roster & Competency Allocation

Engineering allocations align strictly with demonstrated competencies and technical milestones:

Team Member	Discipline	Key Delivered Engineering Artifacts	Week 3 Sprint Status
Blessing Edmund Kwame Dogbe	DevOps / Backend Lead	Render staging deployment, GitHub Actions PR CI automation, mNotify SMS sandbox, production API fallback, release governance.	Staging Active • 108 Tests • Gate 3 Lead
Able Kafu Azanda	Backend Lead Architect	Virtual Queue Engine (PR #13), atomic daily tokens (A-01, B-04), Appointment Cancellation & Reschedule API (PR #14), AuditLog model.	DONE (PR #13 & #14)
Emmanuella Lodonu	Backend Foundations	Single-recipient SMS validation safeguards (mnotify.js), patient registration & lookup API (/api/patients), Call-Next SMS trigger.	GATE 2 DONE • SMS DONE
Harry “Ephraim” Nartey	Fullstack Foundations	Frontend HTTP API client integration, Staff Check-In view (S02/S03), Staff Portal Date Picker & Demo Day toggle (StaffPortal.jsx).	CHECK-IN DONE • WALK-IN
Raymond B. Afrani	Frontend Foundations	Complete patient intake wizard (P01-P07) with speakable reference code (YC-XXXX) & emergency triage diversion.	WIRING P18 & P11-P17

2. Updated Trello Board State: Gate 2 Closeout & Gate 3 Execution

As of Tuesday, 15 September 2026, Gate 2 is 100% complete and verified (12/12 cards). Week 3 has 5 out of 9 deliverables merged and operational:

Lifecycle Stage	Card Count	Percentage	Active Deliverables in Stage
Gate 2 Verified (Done)	12 / 12	100.0%	Complete Patient Intake Loop (P01-P07), Concurrency Lock, Staging & CI, Staff Check-In
Week 3 Verified (Done)	5 / 9	55.6%	Virtual Queue Engine (AA), Cancellation API (AA), SMS Safeguards (EL), Staging Fallback (Blessing), Staff Date Picker (HN)
Week 3 In Progress	3 / 9	33.3%	[FE] P18 Queue Status Screen (RA), [FE] P11-P17 Cancellation UI (RA), [FS] Walk-In Intake (HN)
Week 3 Sprint Backlog	1 / 9	11.1%	[LEAD] Gate 3 Feature-Freeze Verification & Submission (Blessing — Scheduled Fri 18 Sep)
Total Active Scope	21 Cards	81.0%	17 of 21 Tasks Across Gate 2 & Gate 3 Officially Verified & Merged into main

Gate 3 Major Breakthrough: Core Queuing & Cancellation Engines Merged!

- Able Kafu Azanda has successfully merged Pull Request #13 (Virtual Queue Engine & Token Progression State Machine) and Pull Request #14 (Appointment Cancellation & Rescheduling API).
- These deliveries unlock non-colliding daily sequence tokens (A-01, B-04), real-time queue position calculation, dynamic wait times ((patients ahead) × 15 min), atomic slot recovery on cancellation, and zero-collision slot swaps during reschedule.
- All 108 unit and integration tests are passing cleanly on the native Node.js test runner.

3. Completed Work Log: 17 Verified Tasks (Gate 2 Closeout + Week 3)

Task Code & Title	Primary Owner	Milestone	Delivered Technical Artifacts
[BE] MongoDB Schema & Models	Able Kafu Azanda	Gate 2 Done	Mongoose models: Patient, Appointment, Clinician, Room, TimeSlot.
[DEVOPS] SMS Provider Sandbox	Blessing Dogbe	Gate 2 Done	Quick SMS v2 API (mnotify.js), Sender ID 'YenCare' approved.
[BE] Patient Registration & Lookup	Emmanuella Lodonu	Gate 2 Done	POST /api/patients, GET /api/patients/:identifier for index & phone.
[FE] Intake & Triage Flow (P01-P02C)	Raymond Afrani	Gate 2 Done	P01 Home, P02 Details, P02B Site, P02C Emergency Gate diversion.
[BE] Appointment Booking Concurrency Lock	Able Kafu Azanda	Gate 2 Done	Unique compound index locking ({ clinicianId, date, startTime }).
[TEST] Double-Booking Prevention Test	Able Kafu Azanda	Gate 2 Done	Concurrency suite asserting 201 Created vs 409 Conflict.
[BE] Booking SMS Dispatch Trigger	Emmanuella Lodonu	Gate 2 Done	sendSms trigger wired into bookAppointment.js on DB write.
[FS] API Client Service Layer	Harry Nartey	Gate 2 Done	Axios service layer (patients.js, appointments.js), loading spinners.
[FE] Doctor & Slot Picker (P02B-P05)	Raymond Afrani	Gate 2 Done	Doctor selector (P03), Date & 30-min slot picker (P04/P05).
[DEVOPS] Render Staging & CI Pipeline	Blessing Dogbe	Gate 2 Done	Render staging, MongoDB Atlas, GitHub Actions PR CI automation.
[FE] Review & Confirm (P06-P07)	Raymond Afrani	Gate 2 Done	P06 Summary Review and P07 Success card with reference code (YC-XXXX).
[FS] Staff Check-In View (S02/S03)	Harry Nartey	Gate 2 Done	Daily appointment roster with 1-click check-in to CHECKED_IN.
[BE] Virtual Queue Engine & State Machine	Able Kafu Azanda	Week 3 Done	queueEngine.js, QueueCounter.js, atomic tokens (A-01, B-04), FIFO call-next. (PR #13)
[BE] Cancellation & Reschedule API	Able Kafu Azanda	Week 3 Done	PATCH cancel & reschedule, atomic slot recovery & swap, AuditLog. (PR #14)
[BE] SMS Single-Recipient Validation	Emmanuella / Blessing	Week 3 Done	Single recipient validation, phone normalization, Call-Next SMS trigger.
[DEVOPS] Staging Fallback Integration	Blessing Dogbe	Week 3 Done	Vite production fallback to Render staging URL; pool connection hardening.
[FS] Staff Date Picker & Demo Toggle	Harry Nartey	Week 3 Done	StaffPortal.jsx date picker, demo day toggle (2026-09-15), queue roster.

4. Active In-Progress Tasks for Gate 3 Feature-Freeze (4 Tasks)

[FE] Virtual Queue Live Status & Progress Screen (P18)	
Status & Milestone	IN PROGRESS (HIGH PRIORITY) Week 3 (Gate 3 Feature-Complete)
Assignee & Support	Primary: Raymond B. Afrani (RA) • Support: Harry Nartey / Able Kafu Azanda
Scope & Goal	Implement Screen P18 (Queue Status) displaying live sequence token (e.g. A-02), calculated wait time, current serving token, and room banners ('You are 2nd in line', 'Proceed to Consultation Room 1'). Auto-polls /api/appointments/:reference/queue-status every 15 seconds.
Dependencies	Blocked by: None (Backend Queue Engine Merged!) • Blocks: Complete patient attendance loop. Target: Wed 16 Sep 5:00 PM.
Acceptance Criteria	• Renders active token and dynamic wait time based on queue velocity. • Displays dynamic status transitions: CHECKED_IN → WAITING → CALLED → COMPLETED. • Pulsing visual alert and auditory chime when status updates to CALLED. • Polls /api/appointments/:reference/queue-status every 15s with graceful error backoff.

[FE] Self-Service Cancellation & Rescheduling Flow (P11–P17)	
Status & Milestone	IN PROGRESS (HIGH PRIORITY) Week 3 (Gate 3 Feature-Complete)
Assignee & Support	Primary: Raymond B. Afrani (RA) • Support: Harry Nartey / Able Kafu Azanda
Scope & Goal	Build Screens P11 (Appointment Details), P12 (Cancel Confirmation dialog), P13 (Cancellation Complete), and P14–P17 (Reschedule calendar and slot picker). Allows students to look up booking via Reference Code + Phone, review booking, cancel, or pick a new slot.
Dependencies	Blocked by: None (Backend Cancellation API Merged!) • Blocks: Self-service lifecycle. Target: Thu 17 Sep 5:00 PM.
Acceptance Criteria	• Student looks up booking via 8-digit reference code (YC-XXXX) and verified Ghana phone. • Two-step cancellation confirmation releases slot and displays cancellation confirmation card. • Reschedule flow reuses calendar time slot picker and confirms new booking slot. • Displays user-friendly error alerts if appointment is already CHECKED_IN or COMPLETED.

[FS] Walk-In Patient Fast-Track Intake & Desk Triage (S01/S02)	
Status & Milestone	IN PROGRESS Week 3 (Gate 3 Feature-Complete)
Assignee & Support	Primary: Harry "Ephraim" Nartey (HN) • Support: Emmanuella Lodonu / Blessing Dogbe
Scope & Goal	Construct Receptionist Walk-In creation modal on staff portal. Allows clinic receptionists to quickly register an unscheduled student (or lookup existing records via index number) and directly insert them into the day's active virtual queue as a walk-in visit.
Dependencies	Blocked by: None (Patient API & Queue Engine Merged!) • Blocks: Hybrid booking/walk-in queue. Target: Thu 17 Sep 5:00 PM.
Acceptance Criteria	• Input modal collects Student Index, Full Name, Phone, and Priority Level (Standard vs Urgent). • Directly creates patient (if new) and injects appointment with status CHECKED_IN immediately. • Assigns walk-in sequence token (e.g. W-03) positioned appropriately in the queue. • Immediate visual feedback on the staff queue dashboard.

[LEAD] Gate 3 Feature-Freeze Verification & Submission	
Status & Milestone	SCHEDULED Week 3 (Official Gate 3 Milestone)
Assignee & Support	Primary: Blessing Edmund Kwame Dogbe (DevOps / Lead) • Support: AmaliTech Coordinators
Scope & Goal	Execute comprehensive Gate 3 feature review: verify patient booking, SMS alerts, virtual queue, staff consultations, cancellations, and walk-ins are 100% operational on staging. Enforce strict code freeze on main. Submit Gate 3 Progress Assessment to AmaliTech by 5:00 PM.
Dependencies	Blocked by: Week 3 deliverables • Blocks: Gate 3 approval & Week 4 Hardening. Deadline: Fri 18 Sep 2026, 5:00 PM.
Acceptance Criteria	- End-to-end multi-role verification passed on staging (Patient + Receptionist + Doctor). - Branch protection enabled on main: feature freeze enacted, only bugfix PRs accepted. - AmaliTech Gate 3 Submission Form completed with architecture diagrams and live staging URL. - Sprint retrospective and Week 4 hardening kickoff meeting conducted.

5. Engineering Pairing & Acceleration Strategy (Next 48 Hours)

Pairing Allocation	Target Deliverable	Focus & Synergies
Able Kafu Azanda + Raymond Afrani	P18 Queue Screen & P11-P17 Cancellation UI	Able assists Raymond with endpoint request payloads, polling error handlers, and state mappings for the virtual queue and cancellation flows.
Emmanuella Lodonu + Harry Narthey	Walk-In Fast-Track Intake (S01/S02)	Emmanuella pairs with Harry on connecting the staff walk-in modal to the patient registration service and verifying SMS notification triggers.
Blessing Dogbe (DevOps Lead)	Staging Verification & Test Governance	Blessing verifies all automated tests on PR CI (maintaining 108/108 passing), monitors Render cloud logs, and manages release governance.

6. Forward Roadmap: Week 4 Hardening & Week 5 Executive Demo

Week 4: Gate 4 (Reviewed, Fixed & Documented — Target: Fri 25 Sep 2026)

Deliverable / Task	Owner	Scope & Done Criteria
[TEST] Concurrency Stress Testing	Able Kafu Azanda	Simulate 100 simultaneous booking & check-in requests/second; assert p95 latency < 200ms with zero collisions.
[SEC] Data Protection & Phone Masking	Emmanuella Lodonu	Mask phone numbers on public displays (+233 24 **** 567); sanitize all inputs against injection.
[FE] Mobile Responsiveness & A11y	Raymond Afrani	Audit touch target sizes (44px+), contrast ratios, and offline caching on low-bandwidth campus cellular connections.
[FS] Network Resilience & Offline UX	Harry Narthey	Implement IndexedDB fallback and reconnection banners for reception desks during temporary Wi-Fi blips.
[DOCS] Architecture Specs & Runbook	Blessing Dogbe	Publish complete OpenAPI catalog, database ER diagrams, and disaster recovery procedures for AmaliTech.

Week 5: Gate 5 (Demo Delivered + Insights Shared — Target: Fri 2 Oct 2026)

Deliverable / Task	Owner	Scope & Done Criteria
[DEMO] End-to-End Simulation Rehearsal	All Team Members	Simulate an entire morning at KNUST Students' Clinic: 5 online bookings, 2 walk-ins, doctor triage, live SMS texts.
[DATA] Clinical Impact Wait-Time Modeling	Able & Emmanuella	Model simulated throughput improvements: quantify how virtual queuing reduces peak waiting room congestion by 65%.
[SLIDES] Executive Pitch Deck & Video	Raymond & Harry	Author final executive pitch deck and produce 3-minute high-definition product walkthrough video.
[LEAD] Final Capstone Defence & Handover	Blessing Dogbe	Lead final presentation to AmaliTech CSR evaluators & KNUST Health Services stakeholders.